

Assisted Diagnosis of Attention-Deficit Hyperactivity Disorder through EEG Bandpower Clustering with Self-Organizing Maps

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Abstract—The electroencephalogram is an attractive clinical tool given its non-invasive nature, its ability to reflect real-time changes in local cortical activity, and the load of objective bioelectrical measurements that can be derived from it. For decades, the electroencephalogram has been successfully used for diagnosing epilepsy and schizophrenia, among other brain disorders. This paper focuses in the design and implementation of a computer-aided diagnostic tool for establishing the likelihood of presence of Attention-Deficit Hyperactivity Disorder in children, out of routine electroencephalographic recordings obtained during a specific visual stimulation protocol. Classical bandpower features from multiple differential recordings are computed and used as features in a classifier built from a cooperative ensemble of labeled self-organizing maps. Classification accuracy of the proposed system is 0.7 ± 0.11 , as estimated from unseen data, a result that points to the idea that such a quantitative diagnostic aid could adequately support the diagnostic task of a clinical expert.

I. INTRODUCTION

The Attention-Deficit Hyperactivity Disorder (ADHD) is one of the most common psychiatric conditions among children in the 6-17 age range, who present varying degrees of impulsiveness, lack of attention and hyperactivity (conditions that change as the children become of age). This disorder interferes with normal child development and, if missed for timely treatment, will preset the patient for problematic personal, social and psychiatric issues, while significantly deteriorating some behavioral and cognitive functionalities [1].

Patients who have been clinically diagnosed with ADHD through standard psychological testing and family interviews, show deficits similar to those of patients with frontal lobe lesions when performing tasks that require continued attention [2]. Recent electroencephalographic (EEG) research has found consistent differences in activity among children with and without ADHD. Those children who have been diagnosed with the disorder typically show excess power in low frequency rhythms (δ , θ) with a corresponding reduction in power for higher frequency bands (α , β), when compared to normal subjects under closed-eyes condition [2]. Moreover, a significantly larger percentage of epileptoid activity has been observed in diagnosed children, being more evident during maneuvers such as hyperventilation or photostimulation, with additional

presence of inter-hemispheric anomalies such as power asymmetry or reduced parieto-temporal and increased fronto-central coherences [3]. Additional reports strengthen the assumption that subjects with ADHD might have an overall reduction of their cortical activation level, supporting the hypothesis of a maturation delay within the frontal neural systems [4], that is, that when facing a demanding cognitive task there is no evidence of the expected increase of high frequency rhythms.

Current ADHD diagnosis is based mainly in observations of children behavior as reported by parents and/or teachers through standardized questionnaires such as DSM-IV and WISC [5], which are highly subjective. The need for an objective and independent evaluation is relevant so that a confident decision about the presence or absence of this disorder could be established. Given that ADHD is indeed the result of central nervous system disfunction [6], and considering that EEG provides a continuous measure of cerebral activity, it is reasonable to think that this bioelectrical potential carries information that could be used within an adequate decision-support tool for objective evaluation. If better diagnosis is realized, better medication schemes or even neural rehabilitation procedures might be devised in the future.

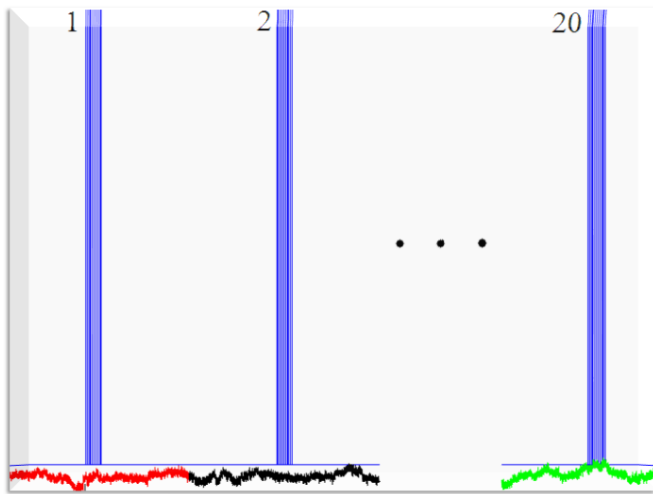
This paper discusses the design and performance of a computational intelligence tool for objective support of the ADHD clinical expert decision. This tool uses bandpower features derived from multi-channel differential EEG recordings as inputs to a self-organizing map (SOM [7]) ensemble, providing a multivariate classification analysis framework. SOM output is presented in a simple visual indication of confidence about healthy or affected conditions.

II. METHODOLOGY

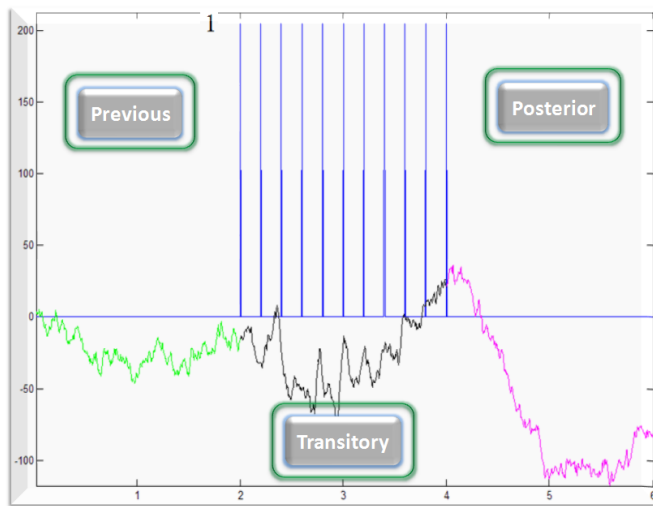
Subjects

Studied population consisted of 28 children with ages between 7-12 years, eighteen of which were diagnosed with ADHD (15 male, three female) and ten without ADHD (five male, five female). Subjects were referred by their educational institutions for psychological evaluation and ADHD diagnosis to the Neuropsychology section of the National Rehabilitation Institute in Mexico City. Evaluation

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(a) Segmentation per each of the 20 repetitions of stimulation.



(b) Pre-, trans-, and post-stimulus segments

Fig. 3. Signal segmentation in relation to stimulation.

connections are established, because although they are not physically connected, each one has some influence over its neighbors. Neighbor influence is determined by the application of a function called neighborhood kernel, $h_{ab}(t) = h(r_a, r_b; t)$, where r_a and r_b are the locations in the grid of nodes a and b , and t is the training time coordinate [7,9]. If the map inputs $x_{band} \in \mathbb{R}^{480}$ are stochastic vectors, assumed to be drawn from the probability density function $p(x_{band})$, then, during the training of the SOM, the codebook vectors are moved so that most of them settle in areas where the density of the data is high while fewer are found in areas where low data occurrences exist.

Classifier Architecture

The SOM ensemble designed for classification consists of two levels of 10x10 node SOMs. The first level maps (SOM_δ , SOM_θ , SOM_α and SOM_β) are trained on pre-, trans-, and post- bandpower features for each band. For a

given input vector x_{band} , its corresponding map will produce a single active node, whose coordinates, y_{band} , are arranged into vector $y = \{y_\delta, y_\theta, y_\alpha, y_\beta\}$. These coordinate vectors are used as features for the second-level map, SOM_f (Fig. 4).

The codebook from the trained SOM_f is labeled for classification in a two step procedure. First, the codebook is clustered using the *k-means* algorithm [10], choosing k as the one that yielded the minimum sum of squared errors. This clustering determines groups of nodes with similar codebook vectors. Figure 5 displays SOM_f nodes in a grid, color-coded by cluster. Second, the probability of a cluster representing a positive ADHD case is computed by counting the activations within the cluster when the map is presented with the labeled training set. Two thresholds of probability are arbitrarily chosen to define each cluster as corresponding to high-, medium- or low-confidence of representing positive ADHD cases. The bottom portion of Fig. 5 shows the cluster probabilities sorted from left to right, together with dashed lines dividing the confidence labels.

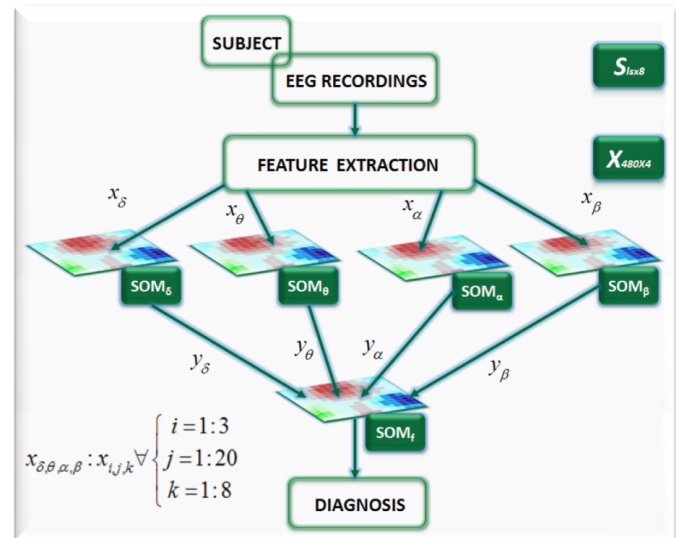


Fig. 4. Diagram for the proposed analysis of the extracted features.

In designing the classifier, a six-fold cross-validation scheme was used, in which the total study group was randomly partitioned into six subgroups each one constituted by three infants with ADHD and two without ADHD. From these subgroups, the training sets sDe (five subgroups) and the validation sets sDv (one subgroup) were formed.

IV. RESULTS

Cross-validation of the designed map structure established that correct classification rate for the system was 0.7 ± 0.1095 . In particular, for an sDe with 22 subjects (14^+ , 8^-) and an sDv with 6 subjects (4^+ , 2^-), seven clusters over the SOM_f were found, defining three distinct regions of probability of positive ADHD in relation to the activated nodes. For example, the yellow cluster includes three activated nodes, two activated for positive ADHD,

raising a confidence probability of 0,66.

Regions of maximum probability were associated to clusters with $p = 1$, moderate probability to those with $p = \frac{2}{3}$ and minimum to those with $p \leq \frac{1}{2}$. Blue and red circles marked over SOM_f in Fig. 5 correspond to nodes activated by sDv (with the subject ID within each circle - total six) and marked as $ADHD^+$ or $ADHD^-$.

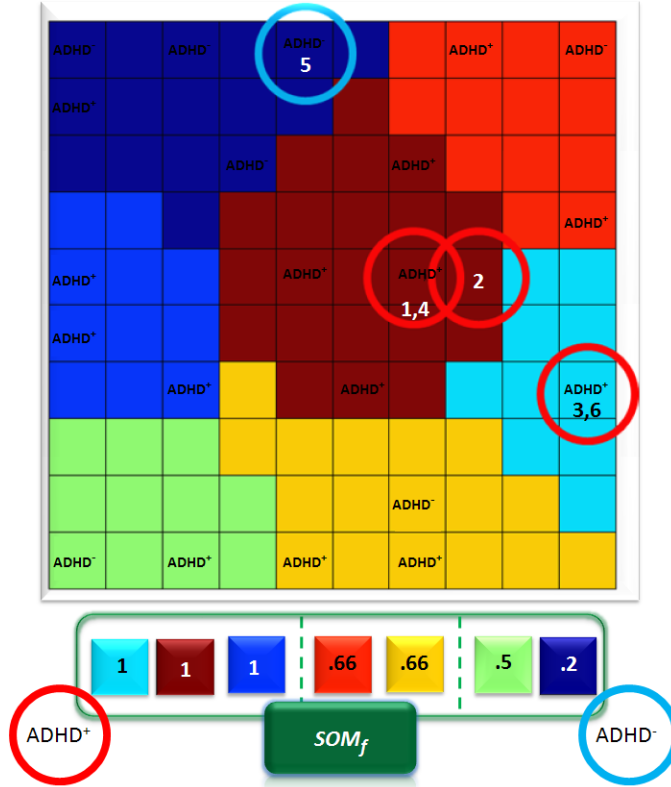


Fig. 5. SOM_f grid, with color-coded clusters. Cluster confidence for positive ADHD representation is indicated in the row of boxes at the bottom of the map; dashed lines separate cluster groups of high-, medium- and low confidence. Circled nodes correspond to outputs of an unseen data set.

V. DISCUSSION AND CONCLUSIONS

Few contributions have proposed quantitative methods for the assessment of ADHD, beyond the studies on EEG power behavior in children with and without this disorder [1-6], so that advances on clinical grounds remain on the subjective, qualitative side. Information obtained from the proposed SOM_f is independent to a large extent of subject collaboration or third-party intervention, which eliminates a major part of the bias immersed in current diagnostic procedures.

The designed tool provides an attractive option to contribute to the body of diagnostic schemes related to the attentional disorder studied. Since for any computer-aided diagnostic tool classification efficiency is dependent on the training set cardinality, it could be expected

that accuracy would be increased as further evidence is collected. It appears that the choice of average power features adequately models the relevant cerebral activity for evaluating the presence of the ADHD, and therefore the SOM ensemble provides confident quantitative support to the clinical evaluator.

VI. ACKNOWLEDGMENTS

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